



GSFA Online

Updated up to the 48th Session of the Codex Alimentarius Commission (2025)

FOOD ADDITIVE DETAILS

Processed eucheuma seaweed (PES) (407a)

Synonym(s)

- PES
- PNG-carrageenan
- semi-refined carrageenan

Functional Classes

- Bulking agent
- Carrier
- Emulsifier
- Gelling agent
- Glazing agent
- Humectant
- Stabilizer
- Thickener

 [Click here to search the FAO JECFA database for the specifications of additive\(s\) with INS No. 407a](#)

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CSV (Tables)

GSFA Provisions for Processed eucheuma seaweed (PES)

Number	Food Category	Max Level	Notes
 14.1.5	Coffee, coffee substitutes, tea, herbal infusions, and other hot cereal and grain beverages, excluding cocoa		GMP  Note 160
 09.2.4.1	Cooked fish and fish products		GMP  Note 325  Note 16
 09.2.4.2	Cooked mollusks, crustaceans, and echinoderms		GMP  Note 16
 06.4.2	Dried pastas and noodles and like products		GMP  Note 256
 01.2.1.2	Fermented milks (plain), heat-treated after fermentation		GMP  Note 234
 01.2.1.1	Fermented milks (plain), not heat-treated after fermentation		GMP  Note 235  Note 234
 04.2.2.7	Fermented vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera) and seaweed products, excluding fermented soybean products of food categories 06.8.6, 06.8.7, 12.9.1, 12.9.2.1 and 12.9.2.3		GMP  Note XS294
 08.1.2	Fresh meat, poultry, and game, comminuted		GMP  Note 281
 08.1.1	Fresh meat, poultry, and game, whole pieces or cuts		GMP  Note 326  Note 16
 06.4.1	Fresh pastas and noodles and like products		GMP  Note 211
 09.2.4.3	Fried fish and fish products, including mollusks, crustaceans, and echinoderms		GMP  Note 41  Note 325  Note 332
 09.2.2	Frozen battered fish, fish fillets, and fish products, including mollusks, crustaceans, and echinoderms		GMP  Note 177  Note 332
 10.2.2	Frozen egg products		GMP
 09.2.1	Frozen fish, fish fillets, and fish products, including mollusks, crustaceans, and echinoderms		GMP  Note XS315  Note XS292

Number	Food Category	Max Level	Notes
			Note XS95 Note 391 Note XS191 Note XS312 Note XS92 Note XS190 Note XS36 Note 332
09.2.3	Frozen minced and creamed fish products, including mollusks, crustaceans, and echinoderms	GMP	
10.2.1	Liquid egg products	GMP	
11.4	Other sugars and syrups (e.g. xylose, maple syrup, sugar toppings)	GMP	Note 258
01.4.1	Pasteurized cream (plain)	GMP	Note XS288
01.2.2	Renneted milk (plain)	GMP	
09.2.5	Smoked, dried, fermented, and/or salted fish and fish products, including mollusks, crustaceans, and echinoderms	GMP	Note 300 Note XS189 Note XS244 Note XS236 Note XS222 Note XS167 Note XS311 Note 332
01.4.2	Sterilized and UHT creams, whipping and whipped creams, and reduced fat creams (plain)	GMP	

Note: Unless otherwise specified, food additive provisions apply to the food category indicated (e.g. Dairy), as well as to all subcategories of that category (e.g. Cheese, Ripened Cheese, etc.).

GSFA Table 3 Provisions

Processed eucheuma seaweed (PES) is a food additive that is included in **Table 3**, and as such may be used in the following foods under the conditions of good manufacturing practices (GMP) as outlined in the Preamble of the Codex GSFA. Although not listed below, Processed eucheuma seaweed (PES) could also be used in heat-treated butter milk of food category 01.1.1 and spices of food category 12.2.1. Note that food categories listed in the **Annex to Table 3** were excluded accordingly. **Processed eucheuma seaweed (PES)** is acceptable in foods conforming to the following commodity standards: CS 70-1981 (for use in packing media only), CS 94-1981 (for use in packing media only), CS 119-1981 (for use in packing media only), CS 249-2006, CS 250-2006, CS 252-2006, CS 273-1968, CS 275-1973, CS 253-2006 (see functional class table and footnote), CS 262-2006 (as anticaking agent only, see functional class table in CXS 262-2006).

Processed eucheuma seaweed (PES) is a: Emulsifier, Stabilizer, Thickener

Any Emulsifier listed in Table 3 can be acceptable for use in all products conforming to CS 117-1981, CS 309R-2011, CS 256-2007.

Any Stabilizer listed in Table 3 can be acceptable for use in all products conforming to CS 117-1981, CS 256-2007.

Any Thickener listed in Table 3 can be acceptable for use in all products conforming to CS 117-1981, CS 256-2007, CS 306-2011.

Number	Food Category
01.1.4	Flavoured fluid milk drinks
01.3	Condensed milk and analogues (plain)
01.4.3	Clotted cream (plain)
01.4.4	Cream analogues
01.5	Milk powder and cream powder and powder analogues (plain)
01.6.1	Unripened cheese
01.6.2	Ripened cheese
01.6.4	Processed cheese
01.6.5	Cheese analogues
01.7	Dairy-based desserts (e.g. pudding, fruit or flavoured yoghurt)
01.8.1	Liquid whey and whey products, excluding whey cheeses
02.2.2	Fat spreads, dairy fat spreads and blended spreads
02.3	Fat emulsions mainly of type oil-in-water, including mixed and/or flavoured products based on fat emulsions
02.4	Fat-based desserts excluding dairy-based dessert products of food category 01.7

Number ▼	Food Category ▼
 03.0	Edible ices, including sherbet and sorbet
 04.1.2	Processed fruit
 04.2.2.2	Dried vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweeds, and nuts and seeds
 04.2.2.3	Vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds in vinegar, oil, brine, or soybean sauce
 04.2.2.4	Canned or bottled (pasteurized) or retort pouch vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds
 04.2.2.5	Vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweed, and nut and seed purees and spreads (e.g., peanut butter)
 04.2.2.6	Vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweed, and nut and seed pulps and preparations (e.g. vegetable desserts and sauces, candied vegetables) other than food category 04.2.2.5
 04.2.2.8	Cooked or fried vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds
 05.0	Confectionery
 06.3	Breakfast cereals, including rolled oats
 06.4.3	Pre-cooked pastas and noodles and like products
 06.5	Cereal and starch based desserts (e.g. rice pudding, tapioca pudding)
 06.6	Batters (e.g. for breading or batters for fish or poultry)
 06.7	Pre-cooked or processed rice products, including rice cakes (Oriental type only)
 06.8	Soybean products (excluding soybean-based seasonings and condiments of food category 12.9)
 07.0	Bakery wares
 08.2	Processed meat, poultry, and game products in whole pieces or cuts
 08.3	Processed comminuted meat, poultry, and game products
 08.4	Edible casings (e.g. sausage casings)
 09.3	Semi-preserved fish and fish products, including mollusks, crustaceans, and echinoderms
 09.4	Fully preserved, including canned or fermented fish and fish products, including mollusks, crustaceans, and echinoderms
 10.2.3	Dried and/or heat coagulated egg products
 10.3	Preserved eggs, including alkaline, salted, and canned eggs
 10.4	Egg-based desserts (e.g. custard)
 11.6	Table-top sweeteners, including those containing high-intensity sweeteners
 12.2.2	Seasonings and condiments
 12.3	Vinegars
 12.4	Mustards
 12.5	Soups and broths
 12.6	Sauces and like products
 12.7	Salads (e.g. macaroni salad, potato salad) and sandwich spreads excluding cocoa- and nut-based spreads of food categories 04.2.2.5 and 05.1.3
 12.8	Yeast and like products
 12.9	Soybean-based seasonings and condiments
 12.10	Protein products other than from soybeans
 13.3	Dietetic foods intended for special medical purposes (excluding products of food category 13.1)
 13.4	Dietetic formulae for slimming purposes and weight reduction
 13.5	Dietetic foods (e.g. supplementary foods for dietary use) excluding products of food categories 13.1 - 13.4 and 13.6
 13.6	Food supplements
 14.1.4	Water-based flavoured drinks, including "sport," "energy," or "electrolyte" drinks and particulated drinks
 14.2.1	Beer and malt beverages
 14.2.2	Cider and perry
 14.2.4	Wines (other than grape)

Number ▾	Food Category ▾
 14.2.5	Mead
 14.2.6	Distilled spirituous beverages containing more than 15% alcohol
 14.2.7	Aromatized alcoholic beverages (e.g. beer, wine and spirituous cooler-type beverages, low alcoholic refreshers)
 15.0	Ready-to-eat savouries
 16.0	Prepared foods

Note: Unless otherwise specified, food additive provisions apply to the food category indicated (e.g. Dairy), as well as to all subcategories of that category (e.g. Cheese, Ripened Cheese, etc.).

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